

## Module 07 Relational (DBMS) Data Access

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### IBM InfoSphere Information Server for Data Integration



## Module Objectives

After completing this topic, you should be able to:

- Describe Connector Stage options and their features
- Describe DB2 Connector Stage
- Use DB2 Connector Stage as a Source
- Use DB2 Connector Stage as a Target
- Describe and use ODBC Connector Stage
- Import table definitions using the Connector Import Wizard and ODBC import

# Connector Options

## ■ ODBC Connector

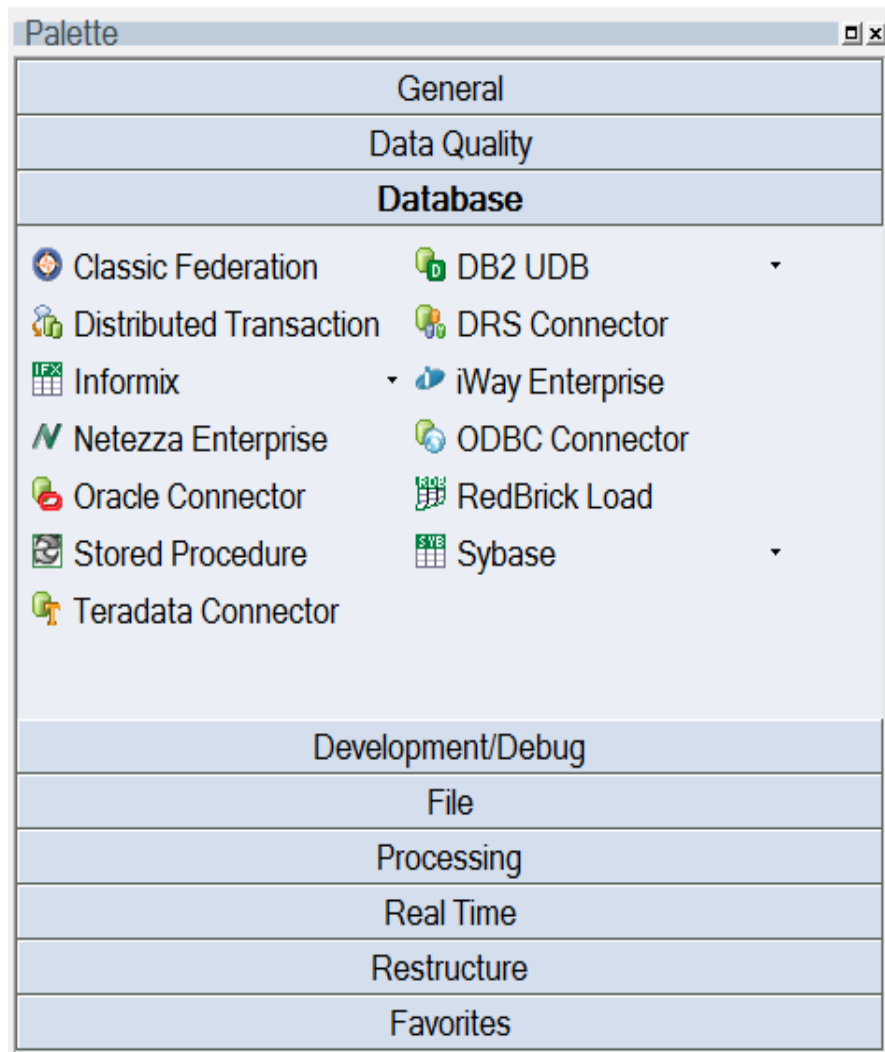
- Extracts from or loads to data sources that support the industry standard Open DataBase Connectivity API
- Need to configure drivers, driver manager, and data source names
  - e.g. ODBC Connector for DB2 UDB, Oracle, Teradata, SQL Server, Informix, flat files, etc.

## ■ RDBMS Connectors

- Offer many features respective to the specific native RDBMS
  - DB2 UDB Connector
  - Oracle Connector
  - Teradata Connector
  - Netezza Connector

## ■ All Connector stages share the same look & feel

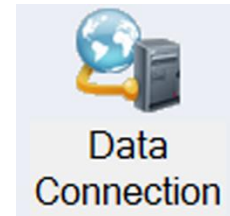
# DataStage - Database Stages



- DB2 Connector
- ODBC Connector
- Teradata Connector
- Oracle Connector
- Sybase Connector
- Netezza Connector
- and many more.....

# Connection Object

- Stores information that is needed to connect to databases in a reusable form
- Quickly define a data source in your connector stages
- Allows for dynamic storage of connection data. E.g. Database authentication passwords must only be changed in one central place. The jobs will pick up the new parameters with each recompile.



| Test Load Save |  |  |
|----------------|--|--|
| SAMPLE         |  |  |
| db2admin       |  |  |
| *****          |  |  |

| Parameter name   | Value                                                                        |
|------------------|------------------------------------------------------------------------------|
| ConnectionString | Type the DSN name                                                            |
| Username         | Type the user name for connecting to the database by using the specified DSN |
| Password         | Type the password for connecting to the database by using the specified DSN. |

## ODBC Connector

- Use ODBC connector in jobs to read data, write data, look up data, and filter data
- Input link exists - The connector writes data
- Output link exists - The connector reads data
- Reject link exists - The connector writes data to this link when data cannot be output to the data source successfully



# Connector Stage - Functionality

- Connection
  - Test connection
  - Load connection property values
- SQL
  - Stage-generated
  - User-defined
  - SQL Builder
- Session
  - Transaction Isolation level
  - Array size
- Before/After SQL
  - SQL statements to be executed before/after any data has been processed
- Write table actions
  - Append
  - Create
  - Replace
  - Truncate
- Write mode
  - Insert, delete, update
  - Insert then update, Update then insert
  - Delete then insert



# Inside the ODBC Connector Stage

The screenshot shows the 'ADDRESS - ODBCConnectorPX' dialog box. It has a 'Link' field set to 'ADDRESS', 'Type' set to 'Output', 'Target stage' set to 'seqAddress', and a 'Variant' dropdown set to '3.5'. Below these are tabs for 'Properties', 'Columns', and 'Advanced'. The 'Properties' tab is active, showing a 'Connection' section with 'Data source' set to 'EMDM\_SRC', and a 'Usage' section with 'Generate SQL' set to 'Yes', 'Table name' set to 'ADDRESS', and 'Enable quoted identifiers' set to 'No'. Other settings include 'SQL' (Select statement), 'Transaction' (Record count: 2000, End of wave: None), and 'Session' (Isolation level: Read uncommitted, Autocommit mode: Off). Annotations with yellow callouts point to the 'Data source' field, the 'Table name' field, and the 'Generate SQL' checkbox.

Select the link or the connector to edit.

Link: ADDRESS  
Type: Output  
Target stage: seqAddress  
Description:  
Variant: 3.5

ADDRESS

Properties | Columns | Advanced

Connection

Data source: EMDM\_SRC  
Username:  
Password:

Usage

Generate SQL: Yes  
Table name: ADDRESS  
Enable quoted identifiers: No

SQL

Select statement:

Enable partitioned reads: No

Transaction

Record count: 2000  
End of wave: None

Session

Isolation level: Read uncommitted  
Autocommit mode: Off

Test Load Save View Data

OK Cancel Help

Data source

Let the stage generate the SQL for this table

Table name



# Columns Tab

The screenshot shows two overlapping windows. The top window is titled "ADDRESS - ODBCConnectorPX" and has a "Columns" tab selected. A yellow callout points to the "Columns" tab. The bottom window is titled "Table Definitions" and shows a tree view with "ADDRESS" selected. A yellow callout points to "ADDRESS". The "Table Definitions" window also has a "Load..." button at the bottom right, with a yellow callout pointing to it.

**Columns tab**

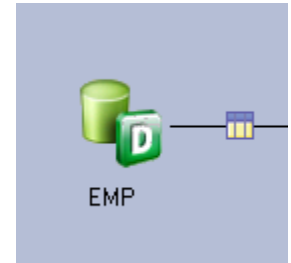
**Select Table Definition**

**Load a Table Definition**

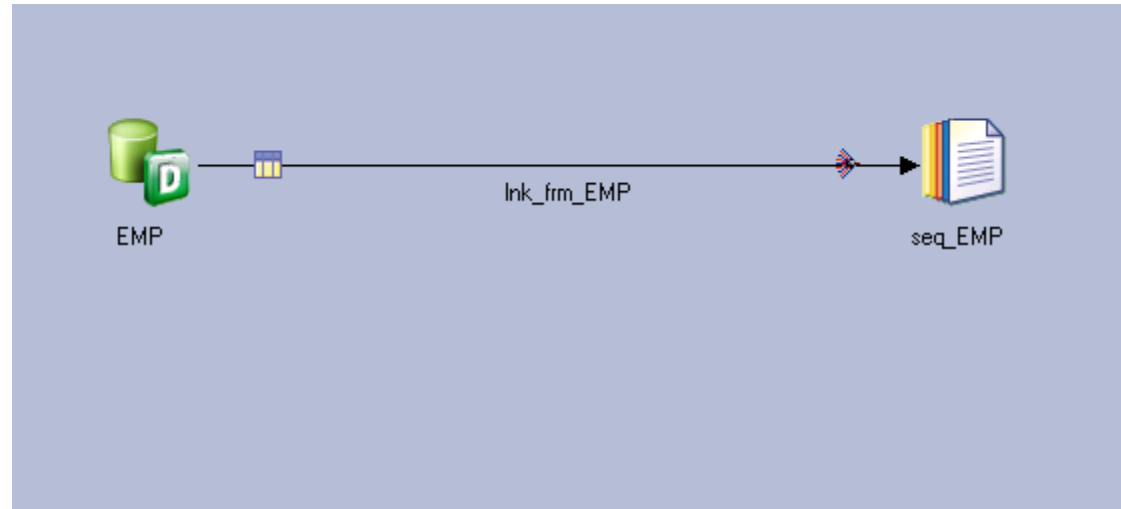
| Column name | Key                      | SQL type | Length | Scale | Nullable | Data element |
|-------------|--------------------------|----------|--------|-------|----------|--------------|
|             | <input type="checkbox"/> | BigInt   | 8      |       | No       |              |
|             | <input type="checkbox"/> | BigInt   | 8      |       | No       |              |
|             | <input type="checkbox"/> | VarChar  | 20     |       | No       |              |
|             | <input type="checkbox"/> | Char     | 1      |       | Yes      |              |
|             | <input type="checkbox"/> | Char     | 1      |       | No       |              |
|             | <input type="checkbox"/> | Char     | 1      |       | Yes      |              |
|             | <input type="checkbox"/> | Char     | 1      |       | Yes      |              |
|             | <input type="checkbox"/> | Char     | 2      |       | Yes      |              |
|             | <input type="checkbox"/> | VarChar  | 100    |       | No       |              |
|             | <input type="checkbox"/> | Char     | 1      |       | Yes      |              |

## DB2 Connector Stage - Overview

- Provides access to DB2 database systems
- Reads, writes, and looks up data in DB2 database tables
- Supports DB2 UDB DPF parallel access
- Performs transactional and non-transactional operations on DB2 data
- Imports metadata of table and column definitions, lists of DB2 database tables, views, and database aliases into a DB2 connector job
- Optionally can have a reject link



## DB2 Connector Stage - As a Source



- Configure a data source connection
- Define usage properties for reading data
- Define column definitions

# DB2 Connector Stage

- Specify connection and usage properties

Ink\_frm\_EMP

Properties Columns Advanced

| ▼ Connection                   |          | Test | Load | Save |
|--------------------------------|----------|------|------|------|
| Instance                       | DB2      |      |      |      |
| Database *                     | SAMPLE   |      |      |      |
| User name                      | db2admin |      |      |      |
| Password                       | xxxxxxxx |      |      |      |
| DB2 client library file        |          |      |      |      |
| ▶ Alternate conductor settings | No       |      |      |      |

| ▼ Usage                         |            | View Data |
|---------------------------------|------------|-----------|
| Generate SQL                    | Yes        |           |
| Table name *                    | SAMPLE.EMP |           |
| Enable quoted identifiers       | No         |           |
| Prefix for expression columns * | EXPR       |           |
| ▼ SQL                           |            |           |
| Select statement *              |            |           |
| ▶ Enable partitioned reads      | No         |           |

## DB2 Connector Stage

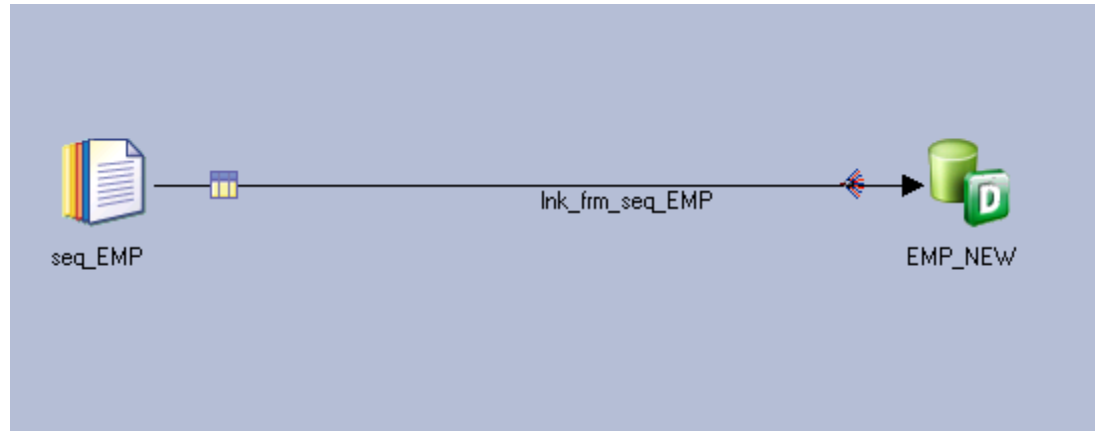
- Specify column definition

Ink\_frm\_EMP

Properties Columns Advanced

|   | Column name | Key                      | SQL type | Length | Scale | Nullable | Data element | Description |
|---|-------------|--------------------------|----------|--------|-------|----------|--------------|-------------|
| 1 | EMPNO       | <input type="checkbox"/> | Integer  | 10     |       | Yes      |              | <none>      |
| 2 | ENAME       | <input type="checkbox"/> | VarChar  | 10     |       | Yes      |              | <none>      |
| 3 | JOB         | <input type="checkbox"/> | VarChar  | 9      |       | Yes      |              | <none>      |
| 4 | MGR         | <input type="checkbox"/> | Integer  | 10     |       | Yes      |              | <none>      |
| 5 | HIREDATE    | <input type="checkbox"/> | Date     | 10     |       | Yes      |              | <none>      |
| 6 | SAL         | <input type="checkbox"/> | Decimal  | 7      | 2     | Yes      |              | <none>      |
| 7 | COMM        | <input type="checkbox"/> | Decimal  | 7      | 2     | Yes      |              | <none>      |
| 8 | DEPTNO      | <input type="checkbox"/> | Integer  | 10     |       | Yes      |              | <none>      |
|   |             | <input type="checkbox"/> |          |        |       |          |              |             |

## DB2 Connector Stage - as a Target



- Configure DB2 Connector stage as a target
- Configure Record Ordering Option if more than one input link
- Define usage properties for writing data
- Define usage properties for bulk loading data
- Partitioning data for a write operation

# DB2 Connector Stage

- Configure DB2 Connector stage as a target

Ink\_frm\_seq\_EMP

Properties Columns Advanced Partitioning

▼ Connection [Test](#) [Load](#) [Save](#)

|                                |            |
|--------------------------------|------------|
| Instance                       | DB2        |
| Database *                     | SAMPLE     |
| User name                      | db2admin   |
| Password                       | XXXXXXXXXX |
| DB2 client library file        |            |
| ▶ Alternate conductor settings | No         |

▼ Usage [View Data](#)

|                                 |                |
|---------------------------------|----------------|
| Write mode *                    | Insert         |
| Generate SQL                    | Yes            |
| Table name *                    | SAMPLE.EMP_NEW |
| Enable quoted identifiers       | No             |
| Prefix for expression columns * | EXPR           |
| ▼ SQL                           |                |
| Insert statement *              |                |



# Sample Teradata Connector

tdcc\_reject\_JT0001FIXD01EV01\_NULL - Teradata Connector

Link:

Type: Input

Source stage: cp1

Description:

Variant:

Properties | Columns | Advanced | Partitioning

**Connection** [Test](#) [Load](#) [Save](#)

|                               |                            |
|-------------------------------|----------------------------|
| Server                        | #MISParams.ETLMIServer#    |
| User name *                   | #MISParams.ETLMISUserid#   |
| Password *                    | #MISParams.ETLMISPassword# |
| Account                       |                            |
| Database                      | #MISParams.ETLMISDB#       |
| Transaction mode              | ANSI                       |
| Client character set          | UTF8                       |
| Automap charset encoding      | Yes                        |
| NLS map name *                |                            |
| Maximum bytes per character * |                            |
| Queryband Expression          |                            |

**Usage** [View Data](#)

|                           |                                          |
|---------------------------|------------------------------------------|
| Write mode *              | Insert                                   |
| Generate SQL              | Yes                                      |
| Table name *              | #MISParams.ETLMISDB#.REJECT_TERM_DEPOSIT |
| Enable quoted identifiers | No                                       |
| Access method             | Immediate                                |
| <b>SQL</b>                |                                          |
| Insert statement *        |                                          |
| Update statement *        |                                          |
| Delete statement *        |                                          |
| ▶ User-defined SQL *      | Statements                               |
| ▶ Table action *          | Append                                   |
| <b>Transaction</b>        |                                          |
| Record count              | 300                                      |
| <b>Session</b>            |                                          |

OK Cancel Help

# Sample Oracle Connector



Link:

Type: Input

Source stage: xdmr\_Prod\_Mark\_Name

Description:

Variant:

Properties | Columns | Advanced | Partitioning

**Connection**

|                             |                                                    |
|-----------------------------|----------------------------------------------------|
| Server                      | #psTDBSecureSiteParamSet.\$TDSecureSiteDBServer#   |
| Username *                  | #psTDBSecureSiteParamSet.\$TDSecureSiteDBUser#     |
| Password *                  | #psTDBSecureSiteParamSet.\$TDSecureSiteDBPassword# |
| Use external authentication | No                                                 |

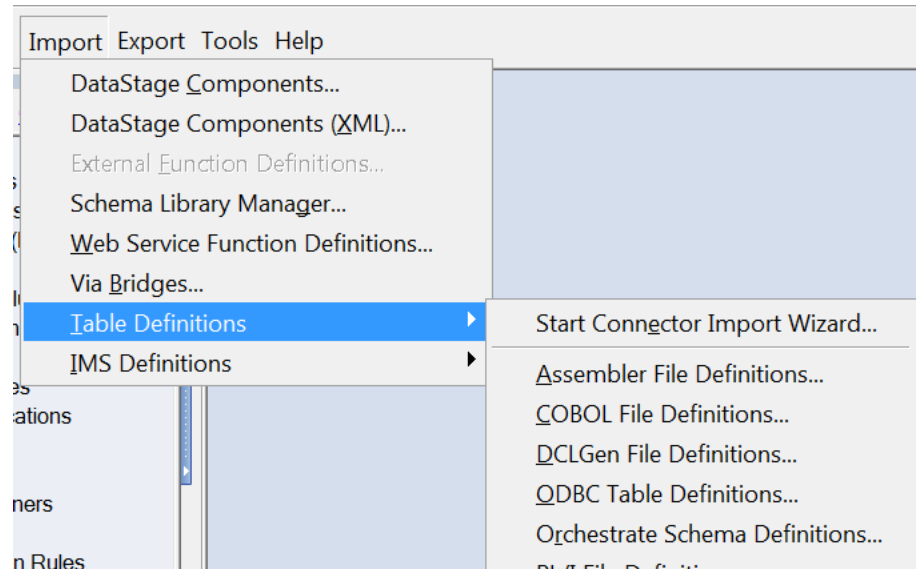
**Usage**

|                                   |                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Write mode                        | Insert                                                                                                                                                 |
| Generate SQL at runtime           | No                                                                                                                                                     |
| Table name *                      |                                                                                                                                                        |
| Enable quoted identifiers         | No                                                                                                                                                     |
| <b>SQL</b>                        |                                                                                                                                                        |
| Insert statement *                | INSERT INTO #psTDBSecureSiteParamSet.\$TDSecureSiteDBSchema# staging_product_marketing_name (CPC, TPC, BRANCHID, LANGUAGE, PRODUCTSHORTNAME, PRODUC... |
| Read insert statement from file   | No                                                                                                                                                     |
| Update statement *                |                                                                                                                                                        |
| Delete statement *                |                                                                                                                                                        |
| PL/SQL block *                    |                                                                                                                                                        |
| Table action *                    | Append                                                                                                                                                 |
| Table name for partitioned writes |                                                                                                                                                        |
| <b>Transaction</b>                |                                                                                                                                                        |
| Isolation level                   | Read committed                                                                                                                                         |

|                                               |                                                                                             |
|-----------------------------------------------|---------------------------------------------------------------------------------------------|
| Array size                                    | 2000                                                                                        |
| Drop unmatched fields                         | No                                                                                          |
| Process warning messages as fatal errors      | No                                                                                          |
| Preserve trailing blanks                      | Yes                                                                                         |
| <b>Logging</b>                                |                                                                                             |
| Log column values on first row error          | No                                                                                          |
| <b>Run before and after SQL statements</b>    | Yes                                                                                         |
| Before SQL statement                          | DELETE FROM #psTDBSecureSiteParamSet.\$TDSecureSiteDBSchema# staging_product_marketing_name |
| Read Before SQL statement from file           | No                                                                                          |
| Fail on error for Before SQL statement        | Yes                                                                                         |
| After SQL statement                           |                                                                                             |
| Read After SQL statement from file            | No                                                                                          |
| Fail on error for After SQL statement         | Yes                                                                                         |
| Before SQL (node) statement                   |                                                                                             |
| Read Before SQL (node) statement from file    | No                                                                                          |
| Fail on error for Before SQL (node) statement | Yes                                                                                         |
| After SQL (node) statement                    |                                                                                             |
| Read After SQL (node) statement from file     | No                                                                                          |
| Fail on error for After SQL (node) statement  | Yes                                                                                         |
| Bulk load options                             |                                                                                             |
| Manage application failover                   | No                                                                                          |

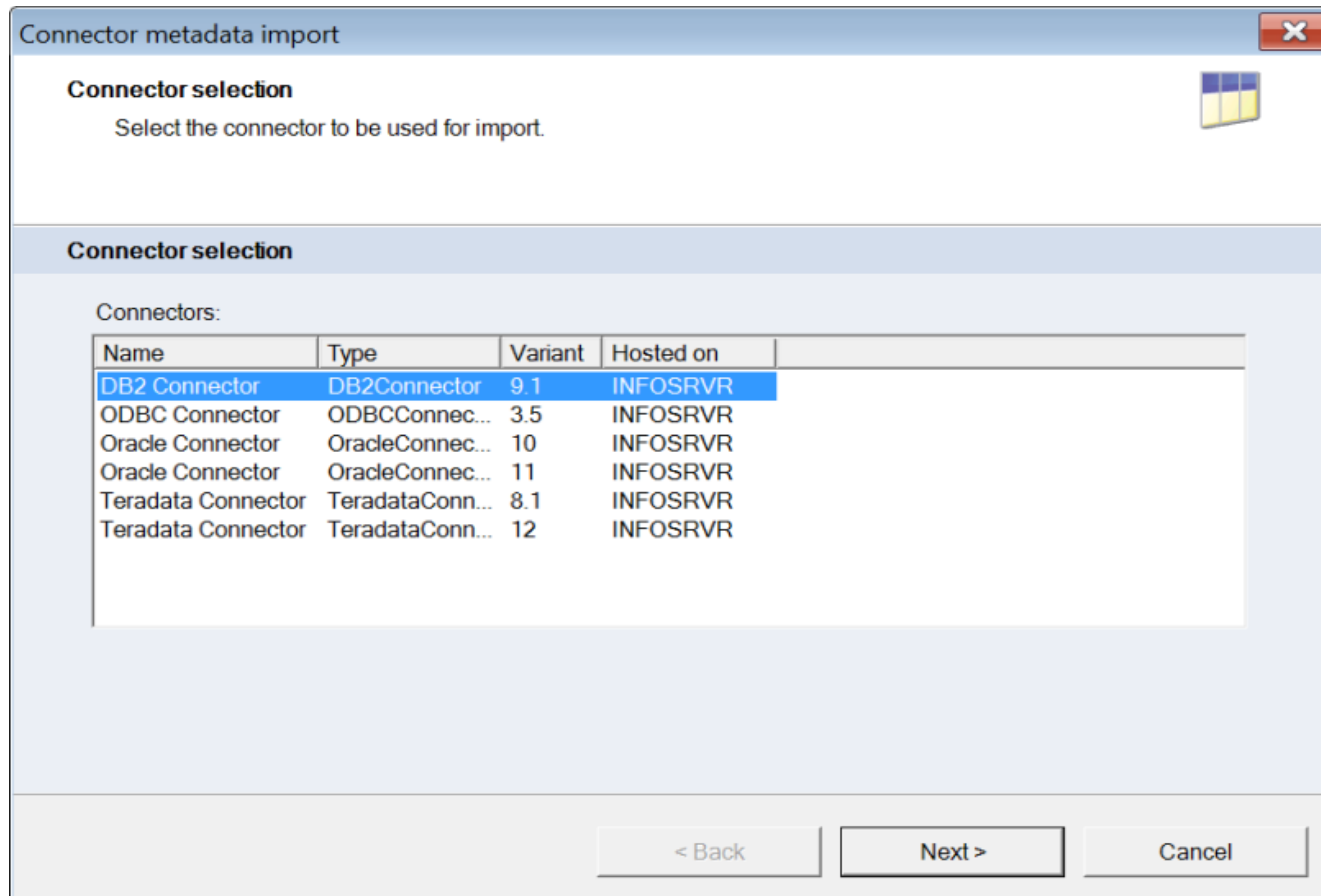
# Importing Table Metadata

- You can import a table definition directly from the source or target database
- Import > Table Definitions > Data Source Type
- Two Import Mechanisms will be featured:
  - Table Metadata Import using the DB2 Connector
  - ODBC Import



# Connector Metadata Import

- Launch from the main menu:
- Import → Table Definitions → Start Connector Import Wizard



# Connector Metadata Import

Connector metadata import

**Connection details**  
Enter the connection details for the import. You can load or save connection details from Data connections defined in t

**Connection** [Test connection](#) [Load...](#) [Save...](#)

Instance  
db2inst1

Database  
SAMPLE

User name  
db2admin

Password  
\*\*\*\*\*

< Back   Next >   Cancel

# Connector Metadata Import

Connector metadata import

**Selection**  
Select the items you wish to import. Certain datasources allow you to view the data or metadata for a given item by

**Selection**

Select

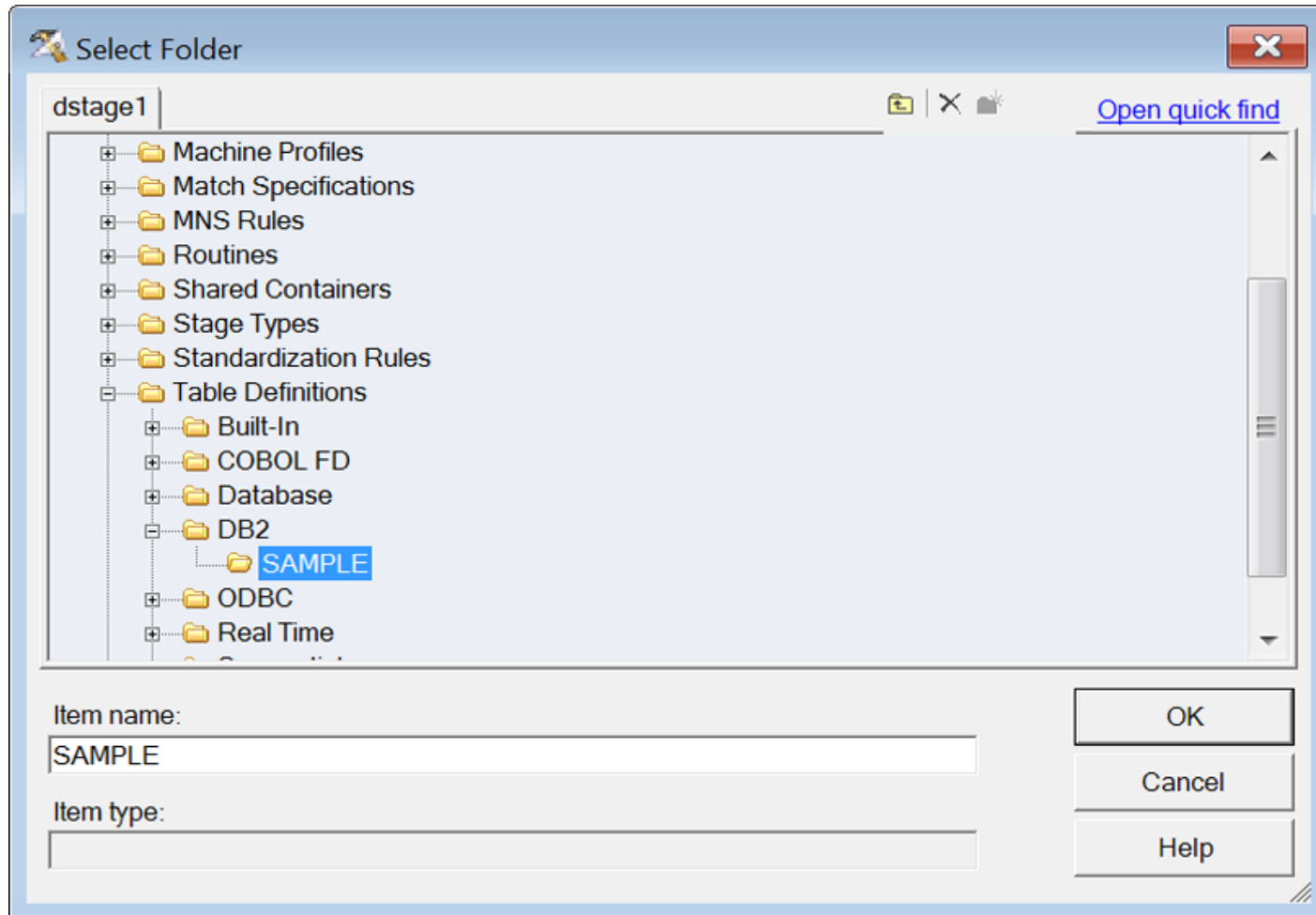
Table or view

BONUS  
DEPT  
EMP  
EMP\_NEW  
EMP\_TEST  
SALGRADE

☐ Include primary keys  
☐ Include foreign keys  
☐ Include indexes  
☐ XML columns as LOBs

Table or View to Import

# Connector Metadata Import



# Import Table Definitions Using ODBC Import

The screenshot shows the 'Import Meta Data (ODBC)' dialog box. It has a title bar with a close button (X). The dialog contains several input fields and buttons. Three yellow callout boxes are present: one pointing to the 'DSN' dropdown menu labeled 'Select DSN Name', one pointing to the 'User name' text box labeled 'Username', and one pointing to the 'Password' text box labeled 'Password'.

**Import Meta Data (ODBC)**

Seen from:  
INFOSRVR

DSN:  
SAMPLE

User name:  
dsadm

Password:  
\*\*\*\*\*

NLS map:  
UTF8

Buttons: OK, Cancel, Help, Filter >>



# Import Table Definitions Using ODBC Import

